Experiment 8: Language Detection and N-gram Generation

A. Language detection using three libraries

```
!pip -q install langdetect textblob langid
from langdetect import detect, DetectorFactory
from textblob import TextBlob
import langid

# Make langdetect results reproducible
DetectorFactory.seed = 0

texts = [
    "Bonjour tout le monde",
    "Hola amigos",
    "Hello everyone"
]

print("1. Language Detection using langdetect")
for text in texts:
    try:
        print(f"{text} -> {detect(text)}")
    except Exception as e:
        print(f"Detection failed for {text!r}: {e}")

print("\n2. Language Detection using TextBlob")
for text in texts:
    try:
        blob = TextBlob(text)

        # Legacy TextBlob versions exposed this method
        if hasattr(blob, "detect_language"):
            print(f"{text} -> {blob.detect_language()}")
        else:
            print(
                f"{text} -> TextBlob language detection "
                "is unavailable in this installed version."
            )
    except Exception as e:
        print(f"TextBlob detection failed: {e}")

print("\n3. Language Detection using langid")
for text in texts:
    try:
        language, score = langid.classify(text)
        print(f"{text} -> {language}, score = {score:.2f}")
    except Exception as e:
        print(f"Detection failed for {text!r}: {e}")
```

```
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
  Building wheel for langdetect (pyproject.toml) ... done
  Building wheel for langid (pyproject.toml) ... done
1. Language Detection using langdetect
Bonjour tout le monde -> fr
Hola amigos -> so
Hello everyone -> no

2. Language Detection using TextBlob
Bonjour tout le monde -> TextBlob language detection is unavailable in this i
Hola amigos -> TextBlob language detection is unavailable in this installed v
Hello everyone -> TextBlob language detection is unavailable in this installe

3. Language Detection using langid
Bonjour tout le monde -> fr, score = -73.45
Hola amigos -> es, score = 0.84
Hello everyone -> it, score = -10.15
```

B. N-gram generation

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.util import ngrams

nltk.download("punkt", quiet=True)
nltk.download("punkt_tab", quiet=True)

text = "Natural language processing enables machines to understand te

tokens = word_tokenize(text)

unigrams = list(ngrams(tokens, 1))
bigrams = list(ngrams(tokens, 2))
trigrams = list(ngrams(tokens, 3))

print("Tokens:", tokens)
print("\nUnigrams:")
print(unigrams)

print("\nBigrams:")
print(bigrams)

print("\nTrigrams:")
print(trigrams)
```

```
Tokens: ['Natural', 'language', 'processing', 'enables', 'machines', 'to', 'u

Unigrams:
[('Natural',), ('language',), ('processing',), ('enables',), ('machines',), (

Bigrams:
[('Natural', 'language'), ('language', 'processing'), ('processing', 'enables

Trigrams:
[('Natural', 'language', 'processing'), ('language', 'processing', 'enables')
```